

STARK: Structure-Aware and Adaptive Representation Learning for Continual Knowledge Graph Embedding

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Abstract

Continual knowledge graph embedding (CKGE) aims to incrementally learn embeddings of entities and relations in a knowledge graph (KG) that evolves over time with a sequence of newly arriving triples. This capability is essential for dynamic applications such as retrieval-augmented generation (RAG), but remains challenging due to the structural diversity and non-uniform growth of evolving KGs. Existing CKGE approaches reveal their limitations in both leveraging rich structural information and achieving update efficiency, as they often rely on simplistic metrics (e.g., degree) and a vanilla TransE loss that is not adaptive to structural dynamics. In this work, we propose *STARK* (Structure-aware and Addaptive Representation learning for CKG), a fast yet effective CKGE framework that enhances structure awareness and supports adaptive optimization. To this end, we propose two major techniques, namely *structural novelty prioritization* (SNP) and *adaptive TransE loss* (ATL). By SNP, STARK allocates higher representational capacity to topologically more important entities, while ATL adaptively keeps embeddings close to the true target of each head-relation pair (i.e., $(h, r, ?)$), dynamically scaling the boundary according to the cardinality of candidate tails. Extensive experiments on multiple CKGE benchmarks demonstrate that STARK achieves both higher accuracy and better efficiency in the time-performance tradeoff than existing state-of-the-art methods. Moreover, embedding visualizations and quantitative analysis confirm that STARK produces more coherent clusters of entities sharing the same $(h, r, ?)$, reflecting improved structural consistency.

CCS Concepts

• Computing methodologies → Knowledge representation and reasoning.

Keywords

Continual graph embedding; Structure-aware; Adaptive loss

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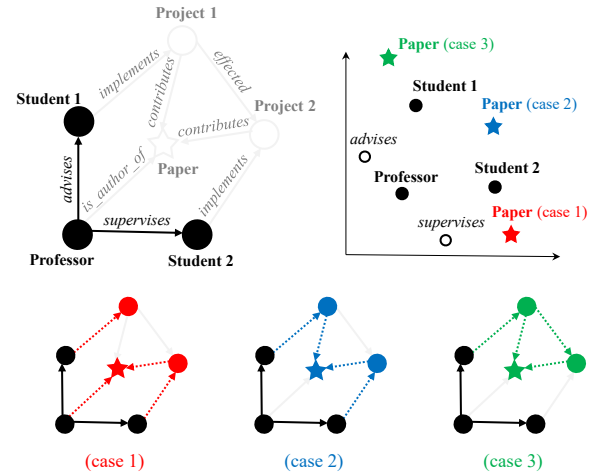


Figure 1: Illustration of structural diversity of continual knowledge graph embedding. As the graph evolves from initial entities (Professor, Student 1, Student 2) to incorporate new ones (Paper, Project 1, Project 2), the position of *Paper* differs greatly across augmentation cases (1–3), yielding large embedding variations despite the same underlying graph.

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1 Introduction

Knowledge graph embedding (KGE) [18, 24] has become a key technique for powering intelligent applications, such as recommendation systems [31], question answering [28, 29], and retrieval-augmented generation (RAG) pipelines [11], all of which rely on up-to-date factual information. In these scenarios, continual knowledge graph embedding (CKGE) is crucial for maintaining high-quality vector representations as new entities and relations are incrementally added to the KG over time. Unlike static KGE methods, CKGE updates embeddings without retraining the entire KG, enabling faster adaptation to knowledge evolution while reducing computational overhead [7, 21].

However, incrementally training over KGs introduces challenges that differ from continual learning in other modalities such as images. First, newly added triples often exhibit structural diversity—for example, some entities emerge as hubs while others remain sparsely connected—leading to non-uniform changes in graph topology [2, 5]. Second, partial updates make it difficult to preserve a globally well-structured embedding space: the model must balance integrating new knowledge and retaining past information, while

avoiding catastrophic forgetting [6, 17]. As illustrated in Figure 1, when new entities such as *Paper* are appended with structurally diverse connection patterns, their embeddings can diverge substantially across augmentation scenarios, showcasing the inherent difficulty of CKGE. Achieving this balance at scale, however, can be computationally demanding.

To boost the training efficiency of CKGE, *FastKGE* [14] was recently proposed, leveraging low-rank adaptation (LoRA) [9] to reduce the size of trainable parameters for newly appended entities and relations. While clearly effective in terms of speed, *FastKGE* still suffers from two limitations: (1) It ignores rich structural information in evolving graphs, relying only on simplistic metrics such as degree or distance—which we empirically show to have negligible impact on embedding quality. (2) It employs the *vanilla* TransE loss [3], which is non-adaptive to structural dynamics in incrementally updated graphs, leading to suboptimal performance when graph topology changes significantly. These limitations motivate the design of a CKGE framework that is both structure-aware and loss-adaptive.

In this work, we propose *STARK* (STRucture-aware and ADAptive Representation learning for CKGE), a fast yet effective CKGE framework designed to be both structure-aware and adaptive to newly arriving entities and relations. Building upon LoRA-based incremental training, *STARK* introduces two novel techniques, namely *structural novelty prioritization* (SNP) and *adaptive TransE loss* (ATL).

LoRA adapters enable efficient training for new entities and relations, but their effectiveness depends on how parameter capacity is distributed among newly appended representations [1, 14]. *STARK* addresses this with SNP, which quantifies entity importance by combining PageRank [4]—capturing global influence—and betweenness centrality [8]—capturing bridging roles across subgraphs. This complementary view of graph structure guides *STARK* to allocate more representational capacity to entities that are both influential and structurally critical in the evolving KG.

ATL, as an adaptive regularization loss, goes beyond the vanilla TransE loss, which relies solely on pairwise comparison between positive and negative entities. It further constrains embeddings to stay around the true target of each head-relation pair, $(h, r, ?)$, yet adaptively adjusts this boundary according to the number of tails sharing the same $(h, r, ?)$. In addition, its influence is scaled by the relative proportion of new entities and relations across snapshots. This design mitigates overly compact clustering in high-cardinality $(h, r, ?)$ groups while improving adaptiveness to structural changes in the evolving KG.

We evaluate *STARK* in the TransE-based CKGE setting, comparing primarily against *FastKGE* [14], the fastest TransE-based incremental embedding approach. Also, we conduct a loss function analysis by replacing our Adaptive TransE Loss (ATL) with other TransE-compatible loss variants such as TransE-RS [33] and TransE-SS [32], to isolate the contribution of ATL to performance improvements. Experimental results on various benchmark datasets demonstrate that:

- **Accuracy:** *STARK* consistently achieves the best MRR and Hit@N across all compared methods, while also outperforming other TransE loss variants.

- **Efficiency:** *STARK* matches the training efficiency of *FastKGE* [14] while delivering higher accuracy.
- **Embedding quality:** Both visualization and quantitative analysis confirm that *STARK* produces more coherent clusters of entities sharing the same (h, r) relation, leading to better structural organization in the embedding space.

2 Related Work

2.1 Continual Knowledge Graph Embedding

Effective CKGE. Continual knowledge graph embedding (CKGE) aims to incrementally update entity and relation embeddings as new triples arrive, without expensive full retraining of the entire KG. With a focus on mitigating catastrophic forgetting, early approaches [6, 7, 21, 25, 27] adapted general continual learning techniques to KGE. For example, Song and Park [21] proposed a continual learning framework to enrich TransE embeddings with newly added facts, while Daruna et al. [7] applied several lifelong learning techniques such as memory replay [16] and regularization [10] to maintain robotic knowledge graphs. These works demonstrated the feasibility of CKGE, but often at high computational cost. More recent methods, such as *IncDE* [13] and *SAGE* [12], introduced specialized strategies to further improve embedding quality, yet they also incur additional overhead from novelty estimation or costly distillation, limiting efficiency under frequent updates.

Efficient CKGE. While earlier CKGE methods emphasized embedding quality and knowledge retention, efficiency has only recently been treated as an explicit design goal. *FastKGE* [14] addresses this by introducing an incremental Low-Rank Adapter (IncLoRA), inspired by LoRA [9] in LLM fine-tuning, to reduce training costs when incorporating new knowledge. Instead of retraining the full embedding, *FastKGE* appends low-rank adapters specifically to new entities and relations, thereby isolating incremental knowledge from the existing KG and alleviating catastrophic forgetting. To mitigate performance loss, it further introduces an adaptive rank allocation heuristic that assigns greater capacity to *important* new entities (e.g., based on degree or distance from the existing KG). While this marks a promising step toward efficient CKGE, its simple importance metrics are limited in capturing structural changes under non-uniform graph growth. Our work follows this efficiency-oriented line of research, but with a focus on more structure-aware prioritization and an adaptive training objective.

2.2 Structural and Adaptive KGE

Structure-Awareness. Structural properties of knowledge graphs have long been exploited in static KGE to improve embedding quality and robustness. For instance, centrality measures such as PageRank [4] or betweenness centrality [8] can identify influential hub nodes that should be embedded distinctly. Prior studies have also investigated the hubness problem in embeddings, which is a phenomenon where a few high-degree entities (hubs) dominate distance computations, causing them to appear spuriously similar to many others. Methods like KBGAN [5] address this by generating harder negative samples, thereby reducing the influence of trivial (often hub-related) negatives, while others propose training adjustments to down-weight frequent entities [2]. However, applying

such structural insights to continual learning remains unexplored. In evolving KGs, certain entities may rapidly emerge as new hubs connecting to many incoming triples, while others remain sparsely connected. This non-uniform growth poses a challenge for parameter allocation and training focus, which our method explicitly aims to address during updates.

KGE Loss Functions. Another largely unexplored aspect in KGE is the use of adaptive loss functions that respond to data characteristics. Most KGE models, from TransE [3] to its variants like TransE-RS [33] and TransE-SS [32], rely on a fixed margin-based ranking loss to compare positive and negative triples [22, 26]. In static settings, a constant margin and uniform negative sampling would work reasonably well. In a continual setting, however, the distribution of new versus old triples can shift over time—for example, negatives involving newly added entities may be easier (if those entities are isolated) or harder (if they form dense clusters) than negatives in the original graph. A fixed margin or weighting scheme cannot capture these dynamics, highlighting the need for loss functions that can adapt to shifting triple distributions in evolving KGs—a challenge that we address in our proposed framework.

3 Methodology

3.1 Preliminaries

CKGE Problem. Let us first formally define the problem of continual knowledge graph embedding (CKGE). A knowledge graph (KG) is represented as $G = (E, R, T)$, where E denotes the set of entities, R the set of relations, and $T = \{(h, r, t)\}$ the set of triples. Each triple $(h, r, t) \in T$ represents that a head entity $h \in E$ is linked to a tail entity $t \in E$ via a relation $r \in R$. Following standard practice, we use the symbol ‘?’ as a wildcard placeholder indicating an unspecified element in a triple (e.g., $(h, r, ?)$ to denote triples sharing a common head–relation pair). In CKGE, the KG evolves over time through a sequence of n increments, denoted as $G_0, \dots, G_{n-1}$, where each increment $G_i = (E_i, R_i, T_i)$ corresponds to the set of newly observed entities, relations, and triples. We use $G^{(i)} = (E^{(i)}, R^{(i)}, T^{(i)})$ to denote the i -th *snapshot* of the cumulative KG up to the i -th increment, i.e., $G^{(i)} = G_0 \cup \dots \cup G_i$. Given $G_0, \dots, G_{n-1}$, the goal of CKGE is to learn d -dimensional embeddings for all entities and relations in $G^{(n-1)}$, namely $\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{R}^d$, such that correct triples (h, r, t) yield smaller distance scores under the TransE formulation:

$$f(h, r, t) = \|\mathbf{h} + \mathbf{r} - \mathbf{t}\|, \quad (1)$$

where a lower value (ideally 0) indicates a more plausible triple, and larger values correspond to structural inconsistency. The key challenge is to incrementally learn embeddings for newly introduced entities and relations without retraining the entire KG from scratch, while maintaining compatibility with previously learned embeddings.

Basic LoRA Framework. Our STARK method builds on the incremental learning framework with low-rank adapters (LoRAs) [14]. Given each i -th increment $G_i = (E_i, R_i, T_i)$, the procedure consists of four steps:

- (1) Estimate the importance of entities in E_i and sort them accordingly.
- (2) Partition the ranked entities into k disjoint groups.

Table 1: Performance comparison of different entity ranking strategies on FB-CKGE with varying numbers of entity groups. Mean reciprocal rank (higher is better) is reported.

Groups (k)	FB-CKGE			
	Dist→Deg	Deg→Dist	Deg Only	Dist Only
2	0.222	0.223	0.221	0.223
5	0.223	0.222	0.216	0.223
10	0.222	0.223	0.214	0.222
20	0.223	0.224	0.215	0.224

- (3) Assign a LoRA module to each group, where the module’s capacity is determined by its rank, and the rank is scaled to the group’s aggregated importance.
- (4) Train the LoRA modules for the new entity groups using the TransE loss (Eq. (5)).

In this way, more representational capacity (i.e., higher LoRA rank) is allocated to entities deemed more important, ensuring that limited parameters are concentrated where they matter most. The key question, however, is how to identify important nodes in the context of CKGE. The existing approach [14] addresses this using simple heuristics such as distance or degree, yet our empirical study demonstrates that these measures contribute little to the final performance.

Limitation of Existing Heuristics. To assess the impact of naive heuristics (i.e., distance and degree) on performance, we compare different sorting strategies based on various permutations of these metrics: *Dist→Deg*, *Deg→Dist*, *Dist Only*, and *Deg Only*. As shown in Table 1, none of these strategies leads to meaningful improvements; in fact, the results are nearly indistinguishable, with mean reciprocal rank (MRR, higher is better) confined to a narrow range (0.222–0.224) across different number of groups ($k \in \{2, 5, 10, 20\}$). Moreover, the *Deg Only* strategy even degrades performance with more groups (i.e., more fine-grained ranking). These findings indicate that degree and distance are insufficient to identify truly impactful new entities. We therefore argue that CKGE requires a more comprehensive notion of structural importance that reflects the broader role of entities in an evolving graph, which motivates the design of our framework STARK.

3.2 Structural Novelty Prioritization

To overcome the limitations of heuristic metrics, STARK introduces structural novelty prioritization (SNP) for entity ranking. For intuition, consider the example in Figure 2, which illustrates a subgraph of the real KG. In terms of degree, the node *Jason Kidd* appears unimportant, being ranked 2763rd by degree. However, from a structural perspective, *Jason Kidd* plays a critical role by directly connecting highly significant neighboring entities such as *marriage*, *San Francisco*, and *UC Berkeley*, despite its low degree. Thus, in terms of connectivity, entities that act as *hubs* or *bridges* linking important concept groups should be regarded as important, regardless of their degree or distance from the existing KG.

SNP Metric. SNP defines a structure-aware importance metric, called the SNP metric, which prioritizes structurally impactful entities and allocates them greater representational capacity (i.e., higher

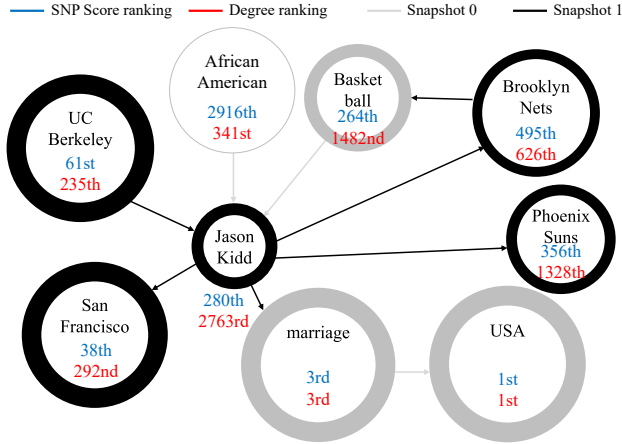


Figure 2: Example subgraph from the ENTITY dataset showing the expansion from snapshot 0 (gray) to snapshot 1 (gray and black). Red numbers indicate degree rankings and blue numbers indicate rankings by SNP. Node diameter is scaled by degree ranking, and its thickness by SNP ranking.

LoRA rank). The SNP metric is designed to capture two essential structural properties: (i) *structural brokerage* and (ii) *global prestige*.

(i) *Structural Brokerage*. Given the i -th snapshot $G^{(i)}$, we aim to assess how much each entity in $E^{(i)}$ functions as a semantic bridge linking otherwise distant concept groups. To this end, we quantify structural brokerage at the *link level*, rather than directly at the entity level. For each direct link (u, v) induced by a triple $(u, ?, v)$ or $(v, ?, u)$ in $G^{(i)}$, we compute its betweenness centrality [8]:

$$BC(u, v) = \sum_{s \neq t} \frac{\sigma_{st}(u, v)}{\sigma_{st}}, \quad (2)$$

where σ_{st} is the number of shortest paths between entities s and t , and $\sigma_{st}(u, v)$ counts those passing through (u, v) .

(ii) *Global Prestige*. We further capture each entity's prominence in the entire graph by applying PageRank [4] on $G^{(i)}$, which reflects how strongly an entity is supported by the importance of its neighbors. For an entity $u \in E^{(i)}$, its PageRank score $PR(u)$ is defined as:

$$PR(u) = (1 - \delta) + \delta \cdot \sum_{v \in N(u)} \frac{PR(v)}{|N(v)|}, \quad (3)$$

where δ is the damping factor, and $N(u)$ is the set of neighboring entities of u through some relation r (i.e., tails of $(u, r, ?)$ or heads of $(?, r, u)$). These scores are computed using the standard power iteration procedure until convergence.

SNP Importance. Finally, we define our importance score $I(u)$ by combining two measures above using a weighted PageRank, where each link between entities u and v is weighted by $BC(u, v)$, thereby modulating PageRank propagation by structural brokerage:

$$I(u) = (1 - \delta) + \delta \cdot \sum_{v \in N(u)} \frac{BC(u, v)}{\sum_{z \in N(v)} BC(v, z)} I(v). \quad (4)$$

This score prioritizes entities that are not only widely referenced but also serve as semantic bridges in the evolving KG. In this way, the resulting distribution emphasizes not only high-degree neighbors but also connections along structurally critical paths. In effect, entities joining around structural hubs (e.g., *Jason Kidd* in Figure 2) stand out with higher importance, while non-central entities are naturally less prioritized, even if they have a high degree (e.g., *African American* in Figure 2).

Grouping and LoRA Allocation. After computing importance scores on $G^{(i)}$, new entities in E_i are sorted by $I(u)$ and partitioned into k disjoint groups $P_1, \dots, P_k$ of equal size. The ordering is defined such that for any $u \in P_i$ and $v \in P_j$ with $i < j$, we have $I(u) \geq I(v)$. We then define the aggregated importance of a group P as: $I(P) = \sum_{u \in P} I(u)$, which serves as a coarse-grained measure of structural significance, reducing sensitivity to noise in individual nodes. Finally, under a fixed average LoRA rank budget, ranks are allocated proportionally to $I(P)$, so that groups with larger importance scores are granted richer parameter subspaces for embedding their entities. In this way, SNP achieves structural adaptiveness in CKGE by aligning parameter allocation with structural novelty.

3.3 Adaptive TransE Loss

Vanilla TransE Loss. The second component of STARK, adaptive TransE loss (ATL), is designed to address a key limitation of the *vanilla* TransE [3] loss in continual settings. Based on the TransE score function $f(h, r, t)$ in Eq. (1), the original TransE loss is defined as:

$$\mathcal{L}_{\text{TransE}} = \sum_{(h, r, t) \in T^+} \sum_{(h', r, t') \in T^-} [\gamma + f(h, r, t) - f(h', r, t')]_+, \quad (5)$$

where $[x]_+$ is $\max(0, x)$, γ is a fixed margin, and T^+ and T^- denote positive and negative triple sets, respectively. T^+ is the set of triples in the original KG, while T^- is constructed by replacement either head h or tail t with a random entity h' or t' . Although this loss enforces relative ordering between positive and random negative triples, its fixed margin cannot accommodate the shifting distribution of entity embedding across snapshots in CKGE. To overcome this, ATL introduces a dynamic margin-based regularization that constrains the absolute score of positive triples $f(h, r, t)$ while adjusting its strength according to the current update's embedding statistics. This allows the model to remain stable under evolving structural distributions.

Dynamic Margin. For each positive triple (h, r, t) , ATL constrains its score $f(h, r, t)$ to remain below a dynamic margin that reflects the structural cardinality of $(h, r, ?)$, defined as:

$$\mathcal{L}_{\text{ATL}} = \sum_{(h, r, t) \in T^+} [f(h, r, t) - \gamma(h, r)]_+, \quad (6)$$

where $\gamma(h, r)$ denotes the dynamic margin. This loss penalizes positive triples whose score exceeds $\gamma(h, r)$. A key problem here is how to dynamically control the margin for each triple. Specifically, we suggest the following margin formulation:

$$\gamma(h, r) = \lambda \cdot \sqrt{n_{h, r}}, \quad (7)$$

where $n_{h, r}$ is the number of distinct tails linked to $(h, r, ?)$ in the current snapshot, and λ is a tunable scaling factor. A larger $n_{h, r}$

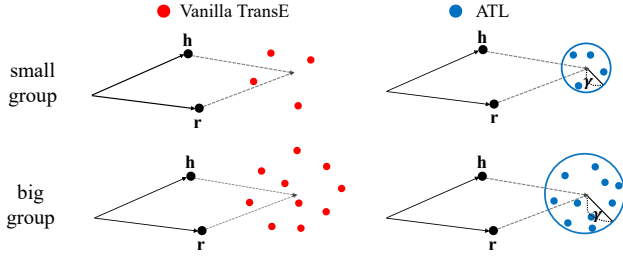


Figure 3: Illustration of Adaptive TransE Loss (ATL). Unlike vanilla TransE which applies a fixed margin, ATL applies a dynamic margin $\gamma(h, r) = \lambda\sqrt{n_{h,r}}$, enforcing tighter constraints for small $(h, r, ?)$ groups and looser ones for large groups, balancing compactness and over-compression.

indicates a higher-cardinality relation, for which a looser constraint avoids forcing many tails into an overly compact region. Conversely, small $(h, r, ?)$ groups receive a tighter bound, ensuring stability in sparse regions. As illustrated in Figure 3, ATL adaptively relaxes or tightens the margin according to group cardinality, preventing embedding collapse while preserving discriminability.

Adaptive Scaling Factor. Another key issue in CKGE settings is how to reflect varying growth patterns of entities or triples across snapshots. To this end, we introduce an adaptive scaling coefficient $\eta^{(i)}$ that controls the overall strength of dynamic margin regularization $\mathcal{L}_{ATL}$ for each increment G_i :

$$\eta^{(i)} = \alpha \cdot \frac{|E^{(i-1)}|}{|E^{(i-1)}| + |E_i|} + (1 - \alpha) \cdot \frac{|T_i|}{|T^{(i-1)}| + |T_i|}, \quad (8)$$

where $\alpha \in [0, 1]$ is a hyperparameter that balances the contributions of entity-based and triple-based effects. Note that we apply asymmetric scaling policies:

- **Entity growth (inverse):** a surge of new entities risks destabilizing embeddings, so η decreases to reduce regularization strength.
- **Triple growth (direct):** more triples provide reliable structural evidence, so η increases accordingly.

This formulation enables ATL to strengthen its influence when relational evidence accumulates, while softening it under rapid entity expansion, achieving stable adaptation across snapshots.

Final Loss. Combining all the above Eqs (5)-(8), our final loss function, adaptive to each i -th snapshot, is defined as:

$$\mathcal{L}^{(i)} = \mathcal{L}_{\text{TransE}} + \eta^{(i)} \cdot \mathcal{L}_{ATL}. \quad (9)$$

This combination allows STARK to retain the efficiency of TransE while improving robustness to diverse growth patterns.

3.4 Overall Framework

For each increment $G_i = (E_i, R_i, T_i)$, the overall workflow of STARK is presented in Algorithm 1. Given the set of triples T_i , we first identify newly introduced entities and relations and update the cumulative snapshot by inserting all new components. Next, we apply structural novelty prioritization (SNP) to adaptively allocate LoRA ranks according to each entity’s structural importance. With these rank allocations, the model is trained using the proposed adaptive TransE loss (ATL), which regularizes positive triples with

Algorithm 1 STARK Training Procedure at the i -th Snapshot

Input: Snapshot $G^{(i-1)}$ and its embeddings $\Theta^{(i-1)}$; current increment G_i
Output: Updated snapshot $G^{(i)}$ and embeddings $\Theta^{(i)}$

- 1: **Entity/Triple Insertion:** Detect newly emerging entities E_i and relations R_i from T_i , constructing $G^{(i)} = (E^{(i)}, R^{(i)}, T^{(i)})$.
- 2: **Structural Importance Estimation:** For each $u \in E^{(i)}$, compute SNP importance score $I(u)$ according to Eq. (4).
- 3: **Rank-Aware Grouping:** Sort E_i by $I(u)$, partition into k groups of equal size, and assign LoRA rank proportionally to group importance $I(P)$.
- 4: **Embedding Optimization:** Train embeddings for G_i by minimizing Eq. (9), obtaining Θ_i .
- 5: **Embedding Aggregation:** Update cumulative embeddings as $\Theta^{(i)} = \Theta^{(i-1)} \cup \Theta_i$.

dynamic margins that reflect the statistical characteristics of the current snapshot.

4 Experiments

4.1 Experimental Setup

Datasets and Metrics. We evaluate STARK on five continual KG benchmarks, all derived from FB15K-237 [20]. Four benchmarks—**ENTITY**, **RELATION**, **HYBRID**, and **FACT**—were introduced by Cui et al. [6], while **FB-CKGE** was proposed by Liu et al. [14]. Although all datasets share the same underlying KG, they differ in how new knowledge is incrementally augmented across snapshots. **ENTITY** emphasizes substantial growth in the number of entities, whereas **RELATION** focuses on the introduction of new relations. **HYBRID** increases both entities and relations simultaneously, while **FACT** introduces relatively small increments of both. In contrast, **FB-CKGE** is designed to mimic real-world scenarios, where each increment integrates only a small portion of new knowledge into a large existing KG; specifically, its initial snapshot already contains about 60% of the triples. Table 2 summarizes the statistics of all datasets. For evaluation, we adopt standard link prediction metrics: mean reciprocal rank (MRR) and Hits@N ($N = 1, 3, 10$).

Baselines. We compare STARK against widely used continual learning methods and the most relevant CKGE baseline. From the continual learning literature, we include incremental-parameter approaches such as progressive neural networks (PNN) [19] and Copy-weight with re-init (CWR) [15], as well as full-parameter fine-tuning methods. The latter group covers replay-based approaches GEM [16] and EMR [23], together with regularization-based methods SI [30] and EWC [10]. These methods serve as general continual learning references but are not specifically tailored for CKGE. For CKGE-specific comparison, we adopt FastKGE [14], which leverages incremental LoRA to achieve parameter-efficient updates. FastKGE has been shown to deliver the best trade-off between accuracy and efficiency among CKGE models. Since it is also TransE-based, FastKGE provides the most direct and fair comparison to STARK, and thus serves as our primary baseline.

Implementation Details. Unless otherwise noted, we follow the basic training settings of FastKGE [14]. All models are optimized with Adam; the embedding dimension d is set to 200, the mini-batch size to 1024, and the negative sampling ratio to 10. For the optimizer,

Table 2: Statistics of the continual KG benchmarks. All datasets converge to the same underlying KG (i.e., $G = G^{(4)}$), but differ in how new entities (E_i), relations (R_i), and triples (T_i) are introduced across snapshots.

Dataset	Increment 0 (G_0)			Increment 1 (G_1)			Increment 2 (G_2)			Increment 3 (G_3)			Increment 4 (G_4)			Entire KG ($G = G^{(4)}$)		
	$ E_0 $	$ R_0 $	$ T_0 $	$ E_1 $	$ R_1 $	$ T_1 $	$ E_2 $	$ R_2 $	$ T_2 $	$ E_3 $	$ R_3 $	$ T_3 $	$ E_4 $	$ R_4 $	$ T_4 $	$ E $	$ R $	$ T $
ENTITY	2,909	233	46,388	2,908	3	72,111	2,908	0	73,785	2,908	1	70,506	2,908	0	47,326			
RELATION	11,561	49	98,819	1,783	47	93,535	411	47	66,136	632	47	30,032	154	47	21,594			
FACT	10,513	237	62,024	2,266	0	62,023	807	0	62,023	308	0	62,023	647	0	62,023	14,541	237	310,116
HYBRID	8,631	86	57,561	1,412	16	20,873	2,737	49	88,017	1,615	58	103,339	146	28	40,326			
FB-CKGE	7,505	237	186,070	3,753	0	31,012	1,876	0	31,012	938	0	31,012	469	0	31,010			

we sweep the learning rate over $\{0.1, 0.2, 0.3\}$ and select the best on the validation set. For parameter-efficient updates, we use a LoRA base rank of 200. In ATL, the margin scaling coefficient λ is selected from $\{0.5, 1.0, 2.0\}$ on the validation set, while the adaptive weight η follows Eq. (8). Early stopping is applied with a patience of 5 based on validation MRR. Experiments are run on a single NVIDIA RTX 4070 SUPER GPU (12GB) with CUDA 11.7 and PyTorch 1.13.

4.2 Results and Analysis

Using the five CKGE benchmarks described above, we evaluate STARK in terms of both effectiveness and efficiency. Specifically, we investigate: (i) its performance on standard link prediction metrics to assess embedding quality, (ii) the contribution of each proposed component through detailed ablation studies, (iii) its training efficiency compared with FastKGE, the most competitive prior CKGE method, and (iv) how STARK influences the geometric structure and organization of the embedding space.

4.2.1 Overall Performance on Link Prediction. Table 3 reports the overall link prediction performance across the five CKGE benchmarks. STARK consistently surpasses all prior methods on every dataset, confirming its robustness to diverse graph evolution patterns. The largest gain appears on the ENTITY benchmark, which involves a substantial number of new entities, while datasets with moderate or minimal growth such as FB-CKGE, HYBRID, and FACT show smaller yet stable improvements. These results indicate that STARK adapts effectively to varying degrees of structural change without sacrificing performance in more static conditions.

We attribute these consistent improvements to the complementary roles of our two proposed components. When structural novelty is high, the SNP mechanism likely helps the model allocate its adaptation capacity toward topologically influential entities, mitigating representation drift caused by new nodes. At the same time, the ATL may contribute by adjusting regularization strength according to relation density, preventing over-compression in dense neighborhoods while maintaining stability in sparse regions. Although these effects will be quantitatively examined in the following ablation studies, the trends observed here already suggest that combining structure-aware prioritization with adaptive regularization leads to robust continual embedding updates across heterogeneous KG dynamics.

4.2.2 Ablation and Component Analysis. Table 4 reports the ablation results that disentangle the respective contributions of SNP and ATL. Several important observations can be made.

First, when only ATL is applied (w/o SNP), the performance gain is minimal, and in some benchmarks nearly indistinguishable from the baseline without any structural guidance. This indicates

that ATL alone cannot fully exploit its adaptiveness in the absence of a mechanism to guide parameter allocation. Since ATL adjusts the constraint strength based on $(h, r, ?)$ group size, it relies on an underlying allocation mechanism that directs capacity towards structurally influential entities. Without such guidance, the adaptive margins have limited effect, suggesting that ATL should be viewed as a complementary component rather than a standalone solution.

Second, applying only SNP (w/o ATL) yields clear and consistent improvements across datasets, except for FACT. On ENTITY and RELATION, where many entities or relations emerge, SNP effectively prioritizes structurally important nodes and redistributes LoRA rank accordingly, leading to substantial gains in MRR and Hits@N. By contrast, FACT introduces very few entities and relations, leaving little structural diversity for SNP to exploit. This confirms that SNP is particularly effective under high structural novelty while remaining neutral when the updates are small.

Finally, when both SNP and ATL are applied together (our full STARK framework), the results are consistently the best across all five benchmarks. This demonstrates their *complementary nature*: SNP provides a principled way to allocate capacity toward topologically impactful newcomers, while ATL adaptively regularizes the embedding objective to maintain discriminative balance across dense and sparse $(h, r, ?)$ groups. The synergy between structure-aware allocation (SNP) and adaptive optimization (ATL) allows STARK to handle diverse growth patterns robustly—whether dominated by entity expansion, relation evolution, or both—empirically validating the benefit of integrating both components.

4.2.3 Effectiveness of ATL in CKGE. To further examine the role of our proposed adaptive loss (ATL) in CKGE settings, we compare it against two well-known TransE variants that also modify the margin-ranking loss with minimal additional overhead, namely TransE-RS [33] and TransE-SS [32]. These baselines were chosen for fairness, as they extend vanilla TransE in a lightweight manner without adding new model components or computational cost.

As shown in Table 5, both alternatives perform poorly in continual settings: their results are consistently lower than those of our ATL model and, in many cases, even underperform vanilla TransE. Interestingly, TransE-SS—previously effective on static KGE benchmarks—exhibits substantial degradation once applied to evolving graphs. This observation underscores that loss designs optimized for fixed graph distributions may become counterproductive when structural distributions shift over time.

In contrast, ATL explicitly adapts its margin to the dynamic cardinality of each $(h, r, ?)$ group, tightening constraints for sparse relations while relaxing them for dense ones. This enables the model to maintain embedding compactness where needed and to prevent

Table 3: Link prediction results on five continual KG benchmarks. We report MRR and Hits@N ($N=1,3,10$).

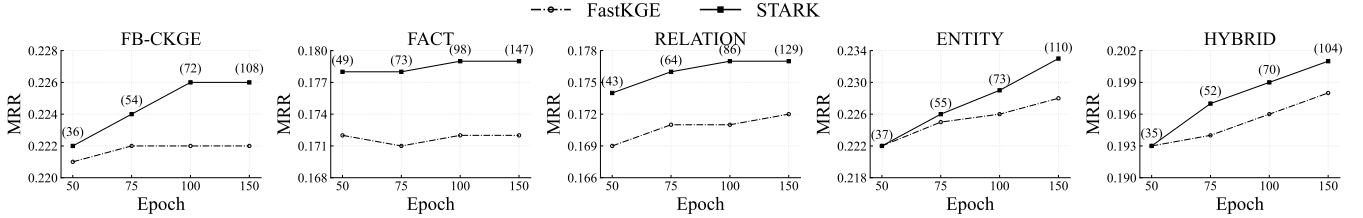
Method	FB-CKGE				FACT				RELATION				ENTITY				HYBRID			
	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10
PNN [19]	0.215	0.122	0.245	0.403	0.157	0.084	0.188	0.290	0.167	0.096	0.191	0.305	<u>0.229</u>	0.130	<u>0.265</u>	0.425	0.185	0.101	0.216	0.349
CWR [15]	0.075	0.011	0.105	0.192	0.083	0.030	0.095	0.192	0.021	0.010	0.024	0.043	0.088	0.028	0.114	0.202	0.037	0.015	0.044	0.077
GEM [16]	0.188	0.103	0.212	0.359	0.175	0.092	0.196	<u>0.345</u>	0.093	0.040	0.106	0.196	0.165	0.085	0.188	0.321	0.136	0.070	0.152	0.263
EMR [23]	0.180	0.097	0.204	0.346	0.171	0.090	0.191	<u>0.337</u>	0.111	0.052	0.126	0.225	0.171	0.090	0.195	0.330	0.141	0.073	0.157	0.267
SI [30]	0.187	0.102	0.211	0.359	0.172	0.088	0.194	0.343	0.113	0.055	0.131	0.224	0.154	0.072	0.179	0.311	0.111	0.049	0.126	0.229
EWC [10]	0.218	0.124	0.247	0.410	0.201	0.113	0.229	0.382	0.165	0.093	0.19	<u>0.306</u>	<u>0.229</u>	0.130	0.264	<u>0.423</u>	0.186	0.102	0.214	<u>0.350</u>
FastKGE [14]	<u>0.222</u>	<u>0.131</u>	<u>0.254</u>	0.402	0.172	0.106	0.197	0.294	<u>0.172</u>	<u>0.103</u>	<u>0.199</u>	0.298	0.228	<u>0.140</u>	0.257	0.398	<u>0.198</u>	<u>0.119</u>	<u>0.230</u>	0.347
STARK (ours)	0.228	0.136	0.261	<u>0.408</u>	<u>0.179</u>	0.113	<u>0.204</u>	0.300	0.178	0.108	0.208	0.308	0.241	0.148	0.273	0.419	0.203	0.123	0.238	0.355

Table 4: Ablation study on two major components of STARK (✓ : included, ✗ : removed).

Method	Components		FB-CKGE				FACT				RELATION				ENTITY				HYBRID			
	SNP	ATL	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10
STARK (ours)	✓	✓	0.228	0.136	0.261	0.408	0.179	0.113	0.204	0.299	0.178	0.108	0.208	0.308	0.241	0.148	0.273	0.419	0.203	0.123	0.238	0.355
w/o ATL	✓	✗	0.225	0.133	0.259	0.408	0.173	0.106	0.200	0.297	0.177	0.107	0.205	0.306	0.239	0.146	0.271	0.418	0.203	0.121	0.237	0.354
w/o SNP	✗	✓	0.221	0.130	0.254	0.396	0.172	0.106	0.197	0.292	0.173	0.104	0.201	0.301	0.231	0.142	0.261	0.406	0.197	0.118	0.229	0.346
w/o SNP & ATL	✗	✗	0.222	0.131	0.254	0.402	0.172	0.106	0.197	0.294	0.172	0.103	0.199	0.298	0.228	0.140	0.257	0.398	0.198	0.119	0.230	0.347

Table 5: Ablation study on TransE loss variants. Results are obtained under identical STARK settings, with only the loss function replaced by TransE-RS [33] or TransE-SS [32].

TransE Loss	FB-CKGE				FACT				RELATION				ENTITY				HYBRID			
	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10
vanilla TransE	0.225	0.133	0.259	0.408	0.173	0.106	0.200	0.297	0.177	0.107	0.205	0.306	0.239	0.146	0.271	0.418	0.203	0.121	0.237	0.354
TransE-RS [33]	0.218	0.128	0.250	0.394	0.175	0.110	0.198	0.294	0.172	0.105	0.200	0.298	0.224	0.138	0.252	0.392	0.196	0.118	0.227	0.342
TransE-SS [32]	0.196	0.116	0.222	0.354	0.166	0.104	0.188	0.280	0.156	0.089	0.184	0.278	0.196	0.126	0.217	0.331	0.175	0.104	0.202	0.308
ATL (ours)	0.228	0.136	0.261	0.408	0.179	0.113	0.204	0.299	0.178	0.108	0.208	0.308	0.241	0.148	0.273	0.419	0.203	0.123	0.238	0.355

**Figure 4: Efficiency comparison between FastKGE and STARK across five continual KG benchmarks. Both methods are trained for the same wall-clock time, and performance is measured by MRR at equivalent time checkpoints. Numbers in parentheses indicate the actual number of epochs trained by STARK within the same runtime budget.**

over-compression in high-density regions, leading to greater robustness under continual graph evolution. Overall, ATL proves not only superior to previous TransE variants but also uniquely suited to continual KGE, where adapting to evolving structures is essential.

4.2.4 Training Efficiency. We evaluate the training efficiency of STARK against FastKGE under an equal wall-clock time budget, as illustrated in Figure 4. For fair comparison, the number of training epochs for each method is normalized by its average per-epoch runtime. The parenthesized numbers next to STARK indicate how many epochs could be completed within the same total training time consumed by FastKGE. A detailed runtime breakdown and calibration procedure are provided in Appendix B.

We focus on FastKGE for efficiency comparison because prior empirical results [14] have shown that other CKGE baselines are significantly slower. Thus, contrasting STARK with FastKGE best isolates the impact of our structure-aware and adaptive design without interference from slower models.

As shown in Figure 4, STARK consistently achieves higher MRR across all five datasets under identical runtime conditions. Even on

the ENTITY benchmark—where training typically demands many more epochs—STARK maintains a clear advantage despite performing fewer updates (i.e., fewer epochs) within the same runtime. These findings demonstrate that STARK achieves a superior efficiency–performance balance: it delivers stronger embeddings than FastKGE while maintaining comparable computational cost, confirming that our structural and adaptive mechanisms introduce minimal overhead relative to their benefits.

4.2.5 Embedding Space Analysis. To gain more insight into how STARK organizes the embedding space, we qualitatively compare it with FastKGE in Figure 5. We visualize the learned high-dimensional embeddings using 2D t-SNE projections to examine their clustering and separation patterns. In the visualization, triangles enclosed by dashed ellipses represent tail entities (the $(h, r, ?)$ groups), while circles enclosed by solid ellipses represent head entities (the $(?, r, t)$ groups). Red and blue colors correspond to the embeddings generated by FastKGE and STARK, respectively.

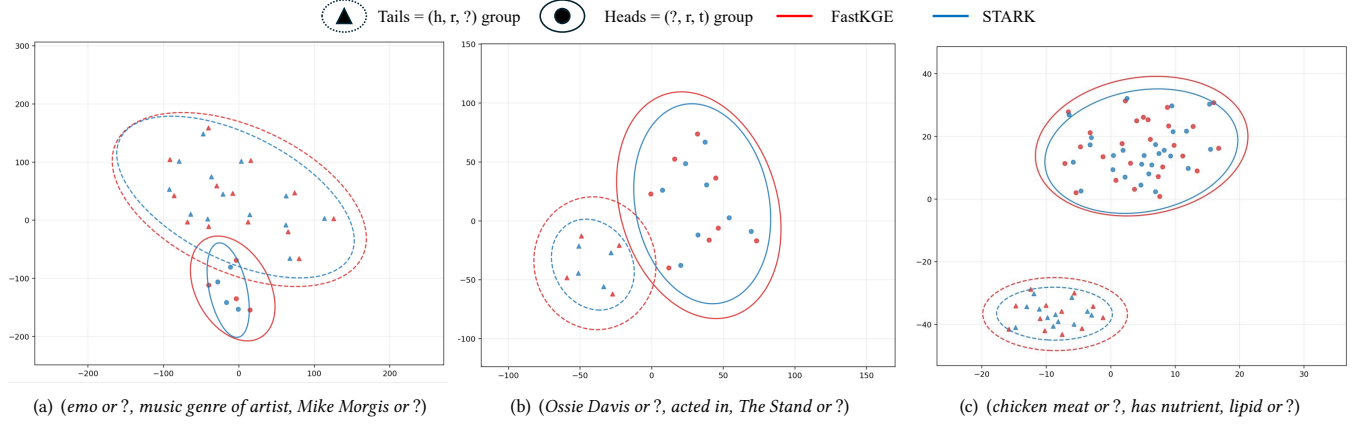


Figure 5: Qualitative embedding analysis with representative examples. Each example is annotated with its corresponding (h, r, t) triple, and the plots show the distribution of tail entities ($(h, r, ?)$ group) and head entities ($(?, r, t)$ group). FastKGE results are shown in red, and STARK results are shown in blue.

Table 6: Quantitative analysis of group compactness using r_{rms} . We report mean $\pm$ standard deviation for both tails ($(h, r, ?)$ groups) and heads ($(?, r, t)$ groups) across all datasets. Lower values indicate more compact embeddings.

	FB-CKGE		FACT		RELATION		ENTITY		HYBRID	
	tails	heads	tails	heads	tails	heads	tails	heads	tails	heads
FastKGE [14]	0.561 $\pm$ 0.200	0.644 $\pm$ 0.182	0.709 $\pm$ 0.172	0.764 $\pm$ 0.146	0.516 $\pm$ 0.262	0.587 $\pm$ 0.489	0.653 $\pm$ 0.287	0.700 $\pm$ 0.250	0.735 $\pm$ 0.398	0.755 $\pm$ 0.334
STARK (ours)	0.502 $\pm$ 0.190	0.582 $\pm$ 0.175	0.562 $\pm$ 0.206	0.650 $\pm$ 0.176	0.413 $\pm$ 0.446	0.489 $\pm$ 0.264	0.548 $\pm$ 0.207	0.604 $\pm$ 0.190	0.656 $\pm$ 0.243	0.696 $\pm$ 0.176

Visualization Examples. In the first case (Figure 5(a), $(h=emo, r=music\ genre\ of\ artist, t=Mike\ Morgis)$), STARK produces a noticeably more compact cluster for both the head group, $(?, music\ genre\ of\ artist, Mike\ Morgis)$, and the tail group, $(emo, music\ genre\ of\ artist, ?)$. Note that such compact clusters correspond to more plausible triples, which yield smaller $f(h, r, t)$ scores (recall Eq. (1)). Similar trends are observed in the second example (Figure 5(b), $(Ossie\ Davis, acted\ in, The\ Stand)$) and the third example (Figure 5(c), $(chicken\ meat, has\ nutrient, lipid)$), where STARK yields less dispersed embeddings. Although ATL explicitly applies its adaptive margin only to the tail groups $(h, r, ?)$, we find that the head groups $(?, r, t)$ also become more compact. This indicates that the adaptive constraint propagates through the shared relation embedding, effectively regularizing both ends of the triple. Since TransE enforces a translation pattern $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$, tightening the tail distribution implicitly narrows the feasible region of the corresponding heads, leading to a more stable geometry in both directions. Furthermore, the observed behavior aligns with ATL’s intended design: smaller groups are tightly concentrated, while larger groups are moderately regularized, consistent with Figure 3. Overall, these visualizations demonstrate that STARK yields more stable clustering in the embedding space compared to FastKGE.

Quantitative Results on Embedding Space. Beyond visualization, we also quantitatively assess the compactness of entity groups in the original d -dimensional space using the root-mean-square radius (r_{rms} ; see Appendix C for details), which measures the average variance per dimension—smaller values indicate tighter clustering. As summarized in Table 6, STARK consistently yields lower r_{rms} than FastKGE across all datasets for both $(h, r, ?)$ and $(?, r, t)$

groups, confirming that the clustering improvements in Figure 5 are systematic rather than incidental.

5 Conclusion

We presented STARK, a structure-aware and adaptive continual knowledge graph embedding framework that unifies structural adaptivity and efficient optimization. STARK integrates two complementary components: structural novelty prioritization (SNP), which identifies topologically important new entities to allocate LoRA capacity, and adaptive TransE loss (ATL), which dynamically adjusts margin constraints based on head-relation pair (h, r) group statistics. Through experiments on five continual knowledge graph benchmarks, STARK consistently outperformed prior methods while maintaining comparable computational cost. Ablation studies confirmed the complementary nature of SNP and ATL, and further analyses showed that our adaptive formulation generalizes better than static TransE variants. Overall, STARK provides a principled and efficient solution for continual knowledge graph embedding, advancing structure-aware adaptation and paving the way for robust reasoning in evolving knowledge environments.

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Appendix

This appendix provides supplementary analyses and additional results that support the main findings reported in the paper. We include detailed ablation studies for both ATL and SNP components, runtime profiling for fair efficiency comparison, and details of quantitative metric of embedding compactness to complement the qualitative visualizations in Section 4.2.5.

A Additional Ablation Studies

A.1 Ablation Study on ATL components

Table A1 analyzes the contribution of different terms in the adaptive coefficient η (Eq. (8)). Recall that η consists of two components: the entity term that down-weights the impact of large numbers of newly introduced entities, and the triple term that emphasizes the influence of new facts. To assess their roles, we conduct ablations by removing each term individually, as well as removing η entirely.

The results show that the full formulation of η , which incorporates both terms, consistently achieves the best performance across all datasets. This indicates that both aspects of graph growth—entities and triples—must be considered simultaneously in order to capture the structural dynamics of continual KGs. Interestingly, on the FACT dataset, eliminating both terms (i.e., removing η) yields slightly higher performance than retaining only one of them. A similar trend is observed in the HYBRID dataset, where relying solely on the triple term even leads to a performance drop. These findings suggest that incomplete modeling of growth patterns may distort the adaptive regularization, while combining both terms allows the model to generalize better to heterogeneous graph evolution.

Table A1: Ablation study on ATL components (✓ : included, ✗ : removed).

Method	ATL Components		FB-CKGE				FACT				RELATION				ENTITY				HYBRID			
	Triple	Entity	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10
STARK (ours)	✓	✓	0.228	0.136	0.261	0.408	0.179	0.113	0.204	0.300	0.178	0.108	0.208	0.308	0.241	0.148	0.273	0.419	0.203	0.123	0.238	0.355
w/o Triple Term (ATL)	✗	✓	0.227	0.135	0.261	0.408	0.173	0.108	0.197	0.292	0.177	0.105	0.208	0.308	0.238	0.145	0.270	0.416	0.203	0.123	0.236	0.353
w/o Entity Term (ATL)	✓	✗	0.227	0.135	0.260	0.406	0.174	0.108	0.198	0.293	0.177	0.106	0.207	0.307	0.234	0.144	0.266	0.409	0.198	0.118	0.231	0.349
w/o Dynamic Coefficient (ATL)	✗	✗	0.221	0.130	0.255	0.396	0.178	0.113	0.202	0.297	0.172	0.105	0.200	0.298	0.234	0.144	0.264	0.407	0.201	0.120	0.235	0.351

Table A2: Ablation study on SNP components (✓ : included, ✗ : removed).

Method	SNP Components		FB-CKGE				FACT				RELATION				ENTITY				HYBRID			
	PageRank	Weighted	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10
STARK (ours)	✓	✓	0.228	0.136	0.261	0.408	0.179	0.113	0.204	0.299	0.178	0.108	0.208	0.308	0.241	0.148	0.273	0.419	0.203	0.123	0.238	0.355
w/o Weighted PR	✓	✗	0.221	0.130	0.256	0.397	0.173	0.107	0.196	0.291	0.175	0.105	0.204	0.303	0.233	0.142	0.263	0.409	0.201	0.121	0.234	0.350
w/o SNP	✗	✗	0.221	0.130	0.254	0.396	0.172	0.106	0.197	0.292	0.173	0.104	0.201	0.301	0.231	0.142	0.261	0.406	0.197	0.118	0.229	0.346

Table A3: Total training time, number of epochs, and average time per epoch for FastKGE and STARK.

Method	FB-CKGE			FACT			RELATION			ENTITY			HYBRID		
	Time	Epochs	Time/Epoch	Time	Epochs	Time/Epoch	Time	Epochs	Time/Epoch	Time	Epochs	Time/Epoch	Time	Epochs	Time/Epoch
FastKGE [14]	1012	180	5.622	2677	150	17.847	1906	149	12.792	1688	180	9.378	1612	150	10.747
STARK (ours)	1291	165	7.824	2466	135	18.267	2964	199	14.894	4564	356	12.820	3259	211	15.445

Table A4: Equivalent epoch alignment for STARK under the same wall-clock time as FastKGE checkpoints.

Method	FB-CKGE Epochs				FACT Epochs				RELATION Epochs				ENTITY Epochs				HYBRID Epochs			
FastKGE [14]	50	75	100	150	50	75	100	150	50	75	100	150	50	75	100	150	50	75	100	150
STARK (ours)	36	54	72	108	49	73	98	147	43	64	86	129	37	55	73	110	35	52	70	104

Overall, this ablation confirms that ATL benefits most from balancing entity- and triple-level growth, highlighting the necessity of jointly modeling both factors in the adaptive coefficient.

A.2 Ablation Study on SNP components

Table A2 analyzes the contribution of the weighting scheme in SNP. In SNP, the importance of new entities is determined using a weighted PageRank algorithm, where the link weights are derived from betweenness centrality. To examine its effectiveness, we compared against two simplified variants: plain PageRank without weighting, and the removal of SNP entirely. The results show that applying plain PageRank brings little or no improvement, whereas adding the proposed betweenness-based weights yields a clear performance boost. This demonstrates that the weighting mechanism is a critical component of SNP, substantially enhancing its ability to identify structurally important entities.

A.3 Ablation on ATL Margin Scaling

To assess the effect of ATL’s adaptive margin, we conduct an ablation study comparing three different scaling strategies: linear (n), logarithmic ($\log n$), and square-root ($\sqrt{n}$). As shown in Table A5, all experiments are performed under identical STARK settings. The results indicate that square-root scaling achieves the best trade-off between adaptivity and stability across all datasets.

B Runtime Profiling and Epoch Calibration

To ensure a fair efficiency comparison between STARK and FastKGE, we first measured the total wall-clock training time and the number of completed epochs for each dataset (Table A3). From these, we computed the average time per epoch, which reflects the intrinsic computational cost of each method. Overall, STARK requires

slightly more time per epoch due to the added structural analysis and adaptive updates.

Based on these measurements, we then aligned the training process by matching wall-clock time rather than raw epochs. Specifically, for each FastKGE checkpoint at 50, 75, 100, and 150 epochs, we calculated the equivalent number of epochs that STARK can complete within the same time budget (Table A4). For example, in FB-CKGE, 50 FastKGE epochs correspond to 36 STARK epochs, and 150 FastKGE epochs correspond to 108 STARK epochs.

This alignment procedure ensures that subsequent comparisons (Figure 4) reflect true efficiency–performance trade-offs under identical time budgets, rather than being biased by per-epoch computational differences.

C Quantitative Metric for Group Compactness

To complement the qualitative embedding visualizations in the main text, we further measure the compactness of entity groups in the original d -dimensional embedding space. Given a set of vectors $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^d$, we compute the sample covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$ as

$$\Sigma = \frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)(x_i - \mu)^\top, \quad \mu = \frac{1}{n} \sum_{i=1}^n x_i. \quad (10)$$

We then measure the root-mean-square radius (r_{rms}). This metric captures the average variance per dimension, defined as:

$$r_{\text{rms}} = \sqrt{\frac{\text{tr}(\Sigma)}{d}}, \quad (11)$$

where $\text{tr}(\Sigma)$ is the trace of Σ (i.e., total variance). A smaller value indicates that the vectors are concentrated more closely around the group mean. As we have shown in Table 6, the (r_{rms}) values are generally lower when STARK is applied. This demonstrates that STARK leads to more compact embeddings by its ATL component.

Table A5: Ablation study on margin scaling variants in ATL. Results are obtained under identical STARK settings, with the margin term replaced by a linear scaling (linear), logarithmic scaling (log), or square-root scaling (sqrt).

Margin Scaling	FB-CKGE				FACT				RELATION				ENTITY				HYBRID			
	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10
linear	0.224	0.132	0.259	0.406	0.171	0.105	0.194	0.292	0.176	0.106	0.204	0.305	0.236	0.145	0.269	0.415	0.198	0.118	0.231	0.348
log	0.225	0.133	0.260	0.407	0.173	0.106	0.198	0.295	0.176	0.106	0.205	0.306	0.237	0.147	0.271	0.416	0.200	0.120	0.232	0.351
sqrt (ours)	0.228	0.136	0.261	0.408	0.179	0.113	0.204	0.300	0.178	0.108	0.208	0.308	0.241	0.148	0.273	0.419	0.203	0.123	0.238	0.355